

To help make virtualization easy, Intel and AMD have developed hardware features which aid in virtualization, namely VT-x and AMD-V by each company. Depending on the make of your processor, if it supports one of these features, virtualization will be smoother and faster. However, this doesn't mean that it will fail otherwise. Along with this, it's better to have a processor with more than one core. While almost every processor has more than one core today, the old ones (up to Pentium 4) didn't have this. While we've seen virtual machines running on a Pentium 4 as well, the performance wasn't all that great.

RAM

When Microsoft says that Windows 7 requires at least 512 MB of RAM for a decent performance, it's obvious that you can't expect good performance when you use Windows 7 as a guest operating system inside Windows Vista (as the host) when you have only 1 GB of RAM. Why? We all know that Vista is a memory hog. It struggles to survive with 1.5 GB of RAM and the performance is really poor. Above that, if you start Windows 7 as a guest, it would take up another 512 MB leaving Vista with only 1 GB and a lot of overhead which will come from the Windows 7 running inside some virtualization software. It's just like trying to run Photoshop's latest version with that much of RAM. While some would say that "Photoshop would run decently", they should know that virtualization software isn't like Photoshop or any other application program because an application program can be fully managed by the operating system. On the other hand, virtualization software run operating systems which demand for hardware more than any application software. If you try to manage the virtual machine on a system with low amount of memory, the virtual machine can get dead slow if you cross the limits of your RAM. The real (host) operating system would try to manage the memory and page out a few things to the disk and then, everything starts getting slower! It's wise to remember that OSes running inside the virtual machine act stubborn and can freeze the machine. The reason for their stubborn behavior is quite simple. The guest OS thinks (well, it can't literally "think" but it's given an environment which makes it believe it can) that all hardware is real while couldn't be further from the truth. The guest operating system tries to take control more rigorously while the virtualization software tries to map the requests of the guest onto the host operating system. All this makes the scene complex. So, always remember to have enough RAM before you launch the virtual machine. In case, you